

cispolyisoprene

## Vulcanization: Cross linking of Rubber

In the year 1839, Charles Good year accidently dropped a mixture of natural rubber and sulphur onto a hot stove. He was surprised to find that the rubber had become strong and elastic. This discovery led to the process that Good year called vulcanization.

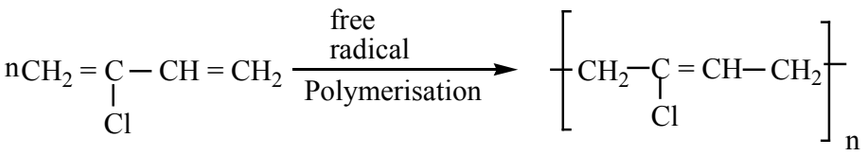
Natural rubber is mixed with 3-5% sulphur and heated at 100-150˚C causes cross linking of the cis-1,4-polyisoprene chains through disulphide (-S-S-) bonds. The physical properties of rubber can be altered by controlling the amount of sulphur that is used for vulcanization. In sulphur rubber, made with about 1 to 3% sulphur is soft and stretchy. When 3 to 10% sulphur is used the resultant rubber is somewhat harder but flexible.

## Synthetic rubber:

Polymerisation of certain organic compounds such as buta-1,3-diene or its derivatives gives rubber like polymer with desirable properties like stretching to a greater extent etc., such polymers are called synthetic rubbers.

## Preparation of Neoprene:

The free radical polymeristion of the monomer, 2-chloro buta-1,3-diene(chloroprene) gives neoprene.

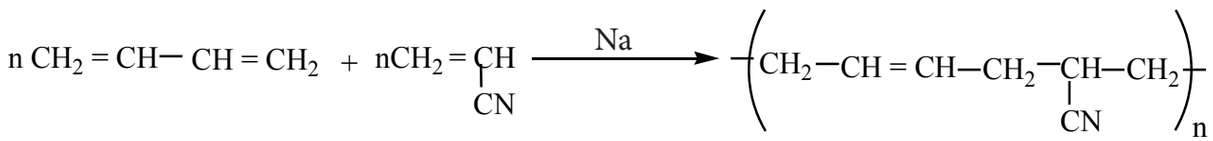



It is superior to rubber and resistant to chemical action.

**Uses:** It is used in the manufacture of chemical containers, conveyer belts.

## Preparation of Buna-N:

It is a co-polymer of acrylonitrile and buta-1,3-diene.




Vinyl cyanide Buna-N

It is used in the manufacture of hoses and tanklinings.

## Preparation of Buna-S:

It is a co-polymer. It is obtained by the polymerisation of buta-1,3-diene and styrene in the ratio 3:1 in the presence of sodium.



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